

Questions used:

There are 5 houses on a street, and each of the first four houses has 3 gnomes in the garden. If there are a total of 20 gnomes on the street, how many gnomes does the fifth house have?

Tina makes \$18.00 an hour. If she works more than 8 hours per shift, she is eligible for overtime, which is paid by your hourly wage + 1/2 your hourly wage. If she works 10 hours every day for 5 days, how much money does she make?

Anna goes trick-or-treating in a subdivision where she gets 14 pieces of candy per house. Her brother Billy goes trick-or-tricking in a neighboring subdivision where he gets 11 pieces of candy per house. If the first subdivision has 60 houses and the second subdivision has 75 houses, how many more pieces of candy does Anna get?

Questions found on [Hugging Face](#)

--- QUESTION 1 ---

1. ZERO-SHOT:

Here's how to solve the problem:

1. **Gnomes in the first four houses:** $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$
2. **Gnomes in the fifth house:** $20 \text{ total gnomes} - 12 \text{ gnomes} = 8 \text{ gnomes}$

Answer: The fifth house has 8 gnomes.

2. FEW-SHOT:

Here's how to solve that:

- * **Gnomes in the first four houses:** $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$
- * **Gnomes in the fifth house:** $20 \text{ total gnomes} - 12 \text{ gnomes} = 8 \text{ gnomes}$

The fifth house has **8** gnomes.

3. CHAIN-OF-THOUGHT:

Here's how to solve the problem step-by-step:

1. **Find the total number of gnomes in the first four houses:**
 - * Each of the first four houses has 3 gnomes.
 - * So, the total gnomes in these four houses is $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$.

2. ****Find the number of gnomes in the fifth house:****

- * There are a total of 20 gnomes on the street.
- * We know 12 of those gnomes are in the first four houses.
- * To find out how many are in the fifth house, subtract the gnomes in the first four houses from the total: $20 \text{ total gnomes} - 12 \text{ gnomes (in first four houses)} = 8 \text{ gnomes}$.

****Answer:**** The fifth house has 8 gnomes.

--- QUESTION 2---

1. ZERO-SHOT:

Here's how to solve Tina's earnings problem step-by-step:

****1. Calculate Regular Pay:****

- * Tina works 8 hours at her regular rate each day.
- * Regular pay per day = 8 hours * \$18.00/hour = \$144.00

****2. Calculate Overtime Hours:****

- * Tina works 10 hours per day, and overtime starts after 8 hours.
- * Overtime hours per day = 10 hours - 8 hours = 2 hours

****3. Calculate Overtime Rate:****

- * Overtime rate = Hourly wage + 1/2 hourly wage
- * Overtime rate = \$18.00 + (1/2 * \$18.00)
- * Overtime rate = \$18.00 + \$9.00 = \$27.00/hour

****4. Calculate Overtime Pay:****

- * Overtime pay per day = Overtime hours * Overtime rate
- * Overtime pay per day = 2 hours * \$27.00/hour = \$54.00

****5. Calculate Total Daily Pay:****

- * Total pay per day = Regular pay per day + Overtime pay per day
- * Total pay per day = \$144.00 + \$54.00 = \$198.00

****6. Calculate Total Earnings for 5 Days:****

- * Total earnings = Total pay per day * Number of days
- * Total earnings = \$198.00/day * 5 days = \$990.00

****Answer:**** Tina makes \$990.00 if she works 10 hours every day for 5 days.

2. FEW-SHOT:

Here's how to solve Tina's earnings:

1. Calculate Regular Pay:

- * **Regular hours per day:** 8 hours
- * **Regular pay per day:** 8 hours * \$18.00/hour = \$144.00
- * **Total regular pay for 5 days:** \$144.00/day * 5 days = \$720.00

2. Calculate Overtime Pay:

- * **Overtime hours per day:** 10 hours (total) - 8 hours (regular) = 2 hours
- * **Overtime hourly rate:** \$18.00 + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00
- * **Overtime pay per day:** 2 hours * \$27.00/hour = \$54.00
- * **Total overtime pay for 5 days:** \$54.00/day * 5 days = \$270.00

3. Calculate Total Earnings:

- * **Total earnings:** \$720.00 (regular pay) + \$270.00 (overtime pay) = \$990.00

Therefore, Tina makes \$990.00.

3. CHAIN-OF-THOUGHT:

Here's how to solve this problem step-by-step:

1. Calculate the regular pay for a single day:

- * Tina works 8 hours at her regular rate.
- * Regular hourly wage = \$18.00
- * Regular pay per day = 8 hours * \$18.00/hour = \$144.00

2. Calculate the overtime hours for a single day:

- * Tina works 10 hours in total.
- * Regular hours = 8 hours
- * Overtime hours per day = 10 hours - 8 hours = 2 hours

3. Calculate the overtime hourly wage:

- * Overtime is paid at her hourly wage + 1/2 her hourly wage.
- * Overtime hourly wage = \$18.00 + (1/2 * \$18.00)
- * Overtime hourly wage = \$18.00 + \$9.00 = \$27.00

****4. Calculate the overtime pay for a single day:****

* Overtime pay per day = Overtime hours * Overtime hourly wage

* Overtime pay per day = 2 hours * \$27.00/hour = \$54.00

****5. Calculate the total pay for a single day:****

* Total pay per day = Regular pay per day + Overtime pay per day

* Total pay per day = \$144.00 + \$54.00 = \$198.00

****6. Calculate the total money Tina makes for 5 days:****

* Tina works for 5 days.

* Total money = Total pay per day * Number of days

* Total money = \$198.00/day * 5 days = \$990.00

****Therefore, Tina makes \$990.00.****

--- QUESTION 3---

1. ZERO-SHOT:

Here's how to solve the problem step-by-step:

****1. Calculate Anna's total candy:****

* Anna gets 14 pieces of candy per house.

* There are 60 houses in her subdivision.

* Total candy for Anna = 14 pieces/house * 60 houses = 840 pieces

****2. Calculate Billy's total candy:****

* Billy gets 11 pieces of candy per house.

* There are 75 houses in his subdivision.

* Total candy for Billy = 11 pieces/house * 75 houses = 825 pieces

****3. Find the difference:****

* To find out how many more pieces Anna gets, subtract Billy's total from Anna's total.

* Difference = Anna's candy - Billy's candy

* Difference = 840 pieces - 825 pieces = 15 pieces

****Answer:** Anna gets 15 more pieces of candy than Billy.**

2. FEW-SHOT:

Here's how to solve the problem:

****1. Calculate Anna's total candy:****

* Anna gets 14 pieces per house * 60 houses = 840 pieces of candy.

****2. Calculate Billy's total candy:****

* Billy gets 11 pieces per house * 75 houses = 825 pieces of candy.

****3. Find the difference:****

* Anna gets 840 pieces - 825 pieces = 15 more pieces of candy.

****Answer:**** Anna gets 15 more pieces of candy.

3. CHAIN-OF-THOUGHT:

Here's how to solve the problem step-by-step:

****1. Calculate the total candy Anna gets:****

* Anna gets 14 pieces of candy per house.

* There are 60 houses in her subdivision.

* Total candy for Anna = 14 pieces/house * 60 houses

****2. Calculate the total candy Billy gets:****

* Billy gets 11 pieces of candy per house.

* There are 75 houses in his subdivision.

* Total candy for Billy = 11 pieces/house * 75 houses

****3. Calculate the difference in candy:****

* We want to find out how many **more** pieces of candy Anna gets.

* Difference = Total candy for Anna - Total candy for Billy

****Now, let's do the calculations:****

* ****Anna's candy:**** 14 * 60 = 840 pieces

* ****Billy's candy:**** 11 * 75 = 825 pieces

* ****Difference:**** 840 - 825 = 15 pieces

****Answer:**** Anna gets 15 more pieces of candy.

--- QUESTION 1 ---

4. SELF-CONSISTENCY (3 Paths):

Path 1: gnomes.

Answer: The fifth house has 8 gnomes.

Path 2: gnomes.

Answer: The fifth house has 8 gnomes.

Path 3: mes

Answer: The fifth house has **8** gnomes.

5. PROMPT CHAINING Final:

Here's how to solve the problem using only the provided variables:

Problem: We need to find out how many gnomes are in the gardens of the houses *after* the first four.

Solution:

1. **Total gnomes on the street:** 20
2. **Gnomes in the first four houses:** 4×3
3. **Gnomes in the houses after the first four:** $20 - (4 \times 3)$

Therefore, the answer is $20 - (4 \times 3)$.

6. GENERATE KNOWLEDGE Final:

Here's how to solve the problem using the provided knowledge:

1. Calculate the total number of gnomes in the first four houses:

* **Formula:** Total gnomes in first four houses = (Number of houses) $\times$ (Gnomes per house)

* **Calculation:** 4 houses $\times$ 3 gnomes/house = 12 gnomes

2. Calculate the number of gnomes in the fifth house:

* **Formula:** Gnomes in the fifth house = (Total gnomes on the street) - (Total gnomes in the first four houses)

* **Calculation:** 20 total gnomes - 12 gnomes = 8 gnomes

Therefore, the fifth house has 8 gnomes.

7. LEAST-TO-MOST Final:

Here's how to solve the problem by answering the sub-questions:

****1. How many gnomes are there in total in the first four houses?****

- * Each of the first four houses has 3 gnomes.
- * To find the total, we multiply the number of houses by the number of gnomes per house: $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$.

****2. What is the combined number of gnomes in the first four houses?****

- * This question is asking for the same information as the first sub-question.
- * The combined number of gnomes in the first four houses is 12 gnomes.

****3. How many gnomes are left for the fifth house after accounting for the gnomes in the first four houses?****

- * We know there are a total of 20 gnomes on the street.
- * We also know that the first four houses have a combined total of 12 gnomes.
- * To find out how many gnomes are left for the fifth house, we subtract the gnomes in the first four houses from the total number of gnomes: $20 \text{ total gnomes} - 12 \text{ gnomes in the first four houses} = 8 \text{ gnomes}$.

****Final Answer:** The fifth house has ****8**** gnomes.**

--- QUESTION 2 ---

4. SELF-CONSISTENCY (3 Paths):

Path 1: \$990.00

****Answer:** Tina makes \$990.00 in 5 days.**

Path 2: earned = \$990.00

****Answer:** Tina makes \$990.00.**

Path 3: ey made = \$990.00

****Answer:** Tina makes \$990.00.**

5. PROMPT CHAINING Final:

Here's how to solve the problem using only the provided variables:

****1. Calculate Overtime Rate:****

- * Overtime rate = Hourly wage + (Hourly wage * Additional fraction for overtime)
- * Overtime rate = $18.00 + (18.00 * 1/2)$
- * Overtime rate = $18.00 + 9.00$

* Overtime rate = 27.00

****2. Determine Overtime Hours per Day:****

* Overtime hours per day = Total hours worked per day - Regular hours per shift

* Overtime hours per day = 10 - 8

* Overtime hours per day = 2

****3. Calculate Daily Overtime Pay:****

* Daily overtime pay = Overtime hours per day * Overtime rate

* Daily overtime pay = 2 * 27.00

* Daily overtime pay = 54.00

****4. Calculate Daily Regular Pay:****

* Daily regular pay = Regular hours per shift * Hourly wage

* Daily regular pay = 8 * 18.00

* Daily regular pay = 144.00

****5. Calculate Total Daily Pay:****

* Total daily pay = Daily regular pay + Daily overtime pay

* Total daily pay = 144.00 + 54.00

* Total daily pay = 198.00

****6. Calculate Total Weekly Pay:****

* Total weekly pay = Total daily pay * Number of days Tina works

* Total weekly pay = 198.00 * 5

* Total weekly pay = 990.00

****Therefore, Tina earns \$990.00 per week.****

6. GENERATE KNOWLEDGE Final:

Here's how to calculate Tina's total earnings for the week, using the provided formulas and her situation:

****Given Information:****

* ****Hourly Wage:**** \$18.00

* ****Regular Hours per Shift:**** 8 hours

* ****Hours Worked per Day:**** 10 hours

* ****Number of Days Worked:**** 5 days

****Calculations:****

****1. Overtime Eligibility Check:****

- * `Hours Worked > Regular Hours`
- * `10 hours > 8 hours` - ****Yes, Tina is eligible for overtime.****

****2. Regular Pay Calculation (per shift):****

- * `Regular Pay = Hourly Wage \* Regular Hours Worked`
- * `Regular Pay = \$18.00 \* 8 hours = \$144.00`

****3. Overtime Hourly Rate Calculation:****

- * `Overtime Hourly Rate = Hourly Wage \* 1.5`
- * `Overtime Hourly Rate = \$18.00 \* 1.5 = \$27.00`

****4. Overtime Pay Calculation (per shift):****

- * First, find the overtime hours worked per shift:
 - * `Overtime Hours Worked = Hours Worked per Day - Regular Hours per Shift`
 - * `Overtime Hours Worked = 10 hours - 8 hours = 2 hours`
- * Now, calculate the overtime pay per shift:
 - * `Overtime Pay = Overtime Hourly Rate \* Overtime Hours Worked`
 - * `Overtime Pay = \$27.00 \* 2 hours = \$54.00`

****5. Total Pay per Shift Calculation:****

- * `Total Pay per Shift = Regular Pay + Overtime Pay`
- * `Total Pay per Shift = \$144.00 + \$54.00 = \$198.00`

****6. Total Earnings for the Week Calculation:****

- * `Total Earnings = Total Pay per Shift \* Number of Days Worked`
- * `Total Earnings = \$198.00 \* 5 days = \$990.00`

****Answer:**** Tina makes ****\$990.00**** for the week.

7. LEAST-TO-MOST Final:

Here's how to solve the problem step-by-step:

****1. What is Tina's regular hourly wage and her overtime hourly wage?****

- * ****Regular hourly wage:**** Tina makes \$18.00 an hour.
- * ****Overtime hourly wage:**** Her overtime is paid at her hourly wage + 1/2 her hourly wage.
 - * 1/2 of her hourly wage = $\$18.00 / 2 = \9.00
 - * Overtime hourly wage = $\$18.00 + \$9.00 = \$27.00$

****Answer to Sub-question 1:**** Tina's regular hourly wage is \$18.00, and her overtime hourly wage is \$27.00.

****2. How many regular hours and how many overtime hours does Tina work in total over the 5 days?****

- * ****Hours per day:**** Tina works 10 hours every day.
- * ****Regular hours per day:**** She is eligible for overtime if she works **more than** 8 hours. So, her first 8 hours are regular hours.
- * ****Overtime hours per day:**** The hours worked beyond the first 8 are overtime hours.
 - * Overtime hours per day = 10 hours - 8 hours = 2 hours.
- * ****Total regular hours over 5 days:****
 - * Total regular hours = 8 hours/day * 5 days = 40 hours.
- * ****Total overtime hours over 5 days:****
 - * Total overtime hours = 2 hours/day * 5 days = 10 hours.

****Answer to Sub-question 2:**** Tina works a total of 40 regular hours and 10 overtime hours over the 5 days.

****3. What is the total amount of money Tina makes from her regular hours and her overtime hours combined?****

- * ****Earnings from regular hours:****
 - * Regular earnings = Total regular hours * Regular hourly wage
 - * Regular earnings = 40 hours * \$18.00/hour = \$720.00
- * ****Earnings from overtime hours:****
 - * Overtime earnings = Total overtime hours * Overtime hourly wage
 - * Overtime earnings = 10 hours * \$27.00/hour = \$270.00
- * ****Total money made:****
 - * Total earnings = Regular earnings + Overtime earnings
 - * Total earnings = \$720.00 + \$270.00 = \$990.00

****Answer to Sub-question 3:**** Tina makes a total of \$990.00 from her regular hours and her overtime hours combined.

****Final Answer:**** Tina makes \$990.00.

--- QUESTION 3 ---

4. SELF-CONSISTENCY (3 Paths):

Path 1: r: Anna gets 15 more pieces of candy than Billy.

Path 2: es

****Answer:**** Anna gets 15 more pieces of candy.

Path 3: r:** Anna gets 15 more pieces of candy than Billy.

5. PROMPT CHAINING Final:

Here's how to solve the problem using only the provided variables:

1. Calculate Anna's total candy:

- * Anna's candy per house: 14
- * Number of houses Anna visits: 60
- * Anna's total candy = $14 * 60$

2. Calculate Billy's total candy:

- * Billy's candy per house: 11
- * Number of houses Billy visits: 75
- * Billy's total candy = $11 * 75$

3. Determine who got more candy:

- * Compare the result of $(14 * 60)$ with the result of $(11 * 75)$.

The solution involves performing these calculations and then comparing the two results.

6. GENERATE KNOWLEDGE Final:

Here's how to solve the problem step-by-step:

1. Calculate Anna's total candy:

- * Anna gets 14 pieces of candy per house.
- * There are 60 houses in her subdivision.
- * Anna's total candy = $14 \text{ pieces/house} * 60 \text{ houses} = 840 \text{ pieces}$

2. Calculate Billy's total candy:

- * Billy gets 11 pieces of candy per house.
- * There are 75 houses in his subdivision.
- * Billy's total candy = $11 \text{ pieces/house} * 75 \text{ houses} = 825 \text{ pieces}$

3. Find the difference in candy:

- * Anna's total candy = 840 pieces
- * Billy's total candy = 825 pieces
- * Difference = $840 \text{ pieces} - 825 \text{ pieces} = 15 \text{ pieces}$

Answer: Anna gets 15 more pieces of candy than Billy.

7. LEAST-TO-MOST Final:

Here's how to solve the problem step-by-step:

****1. How many pieces of candy does Anna get in total?****

- * Anna gets 14 pieces of candy per house.
- * There are 60 houses in her subdivision.
- * Total candy for Anna = (pieces per house) * (number of houses)
- * Total candy for Anna = $14 * 60 = 840$ pieces of candy.

****2. How many pieces of candy does Billy get in total?****

- * Billy gets 11 pieces of candy per house.
- * There are 75 houses in his subdivision.
- * Total candy for Billy = (pieces per house) * (number of houses)
- * Total candy for Billy = $11 * 75 = 825$ pieces of candy.

****3. What is the difference between the total number of pieces of candy Anna gets and the total number of pieces of candy Billy gets?****

- * Anna's total candy = 840 pieces.
- * Billy's total candy = 825 pieces.
- * Difference = Anna's total candy - Billy's total candy
- * Difference = $840 - 825 = 15$ pieces of candy.

****Final Answer:**** Anna gets 15 more pieces of candy.

===== STARTING LOOPS & VERIFICATION =====

--- QUESTION 1 ---

8. SELF-ASK:

Yes.

Follow up: How many gnomes are in the gardens of the first four houses?

Intermediate answer: $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$.

Follow up: How many gnomes are left for the fifth house?

Intermediate answer: $20 \text{ total gnomes} - 12 \text{ gnomes} = 8 \text{ gnomes}$.

So the final answer is: 8 gnomes.

--- QUESTION 2 ---

8. SELF-ASK:

Yes.

Follow up: How much does Tina make per hour for regular hours?

Intermediate answer: \$18.00 per hour.

Follow up: How many regular hours does Tina work per shift?

Intermediate answer: 8 hours.

Follow up: How many overtime hours does Tina work per shift?

Intermediate answer: 10 hours - 8 hours = 2 hours.

Follow up: What is Tina's overtime hourly wage?

Intermediate answer: $\$18.00 + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00$ per hour.

Follow up: How much does Tina make for regular hours per shift?

Intermediate answer: 8 hours * \$18.00/hour = \$144.00.

Follow up: How much does Tina make for overtime hours per shift?

Intermediate answer: 2 hours * \$27.00/hour = \$54.00.

Follow up: How much does Tina make per shift?

Intermediate answer: \$144.00 + \$54.00 = \$198.00.

Follow up: How many days does Tina work?

Intermediate answer: 5 days.

Follow up: What is Tina's total earnings for the week?

Intermediate answer: \$198.00/day * 5 days = \$990.00.

So the final answer is: \$990.00.

--- QUESTION 3 ---

8. SELF-ASK:

Yes.

Follow up: How many pieces of candy does Anna get in total?

Intermediate answer: 14 pieces/house * 60 houses = 840 pieces.

Follow up: How many pieces of candy does Billy get in total?

Intermediate answer: 11 pieces/house * 75 houses = 825 pieces.

Follow up: How many more pieces of candy does Anna get than Billy?

Intermediate answer: 840 - 825 = 15 pieces.

So the final answer is: 15 pieces.

===== STARTING LOOPS & VERIFICATION =====

--- QUESTION 1 ---

9. REFLEXION:

[Draft]: the fifth house: ** 20 total gnomes - 12 gnomes = 8 gnomes

Answer: The fifth house has 8 gnomes....

[Critique]:

Here's a critique of the draft answer:

****Critique:****

The math in the draft answer is ****correct****.

* ****Step 1:**** 4 houses * 3 gnomes/house = 12 gnomes. This calculation accurately determines the total number of gnomes in the first four houses.

* ****Step 2:**** 20 total gnomes - 12 gnomes = 8 gnomes. This calculation correctly subtracts the gnomes from the first four houses from the total number of gnomes to find the number of gnomes in the fifth house.

****Errors:****

There are ****no mathematical errors**** in the draft answer. The logic and calculations are sound.

[Final Revision]:

The final answer is $\boxed{8}$.

10. CHAIN-OF-VERIFICATION:

[Baseline Draft]: the fifth house: **** 20 total gnomes - 12 gnomes = 8 gnomes**

****Answer:**** The fifth house has 8 gnomes....

[Verification Plan]:

Here are two verification questions to check if the math is correct:

1. ****If the fifth house had 8 gnomes, and the first four houses each had 3 gnomes, what would be the total number of gnomes?**** (This checks if the calculation of the total from the individual parts is correct.)

2. ****If there were 20 gnomes in total, and 8 of them were in the fifth house, how many gnomes would be in the remaining four houses combined?**** (This checks if the subtraction step is correct and if the remaining gnomes can be evenly distributed among the first four houses.)

[Verifications Executed]:

Let's break down these verification questions:

****1. If the fifth house had 8 gnomes, and the first four houses each had 3 gnomes, what would be the total number of gnomes?****

* ****Gnomes in the first four houses:**** 4 houses * 3 gnomes/house = 12 gnomes

* ****Gnomes in the fifth house:**** 8 gnomes

* ****Total number of gnomes:**** 12 gnomes + 8 gnomes = ****20 gnomes****

****2. If there were 20 gnomes in total, and 8 of them were in the fifth house, how many gnomes would be in the remaining four houses combined?****

- * **Total gnomes:** 20 gnomes
- * **Gnomes in the fifth house:** 8 gnomes
- * **Gnomes in the remaining four houses:** $20 \text{ gnomes} - 8 \text{ gnomes} = 12 \text{ gnomes}$

Verification of the second part of question 2:

The question also asks if the remaining gnomes can be evenly distributed among the first four houses.

- * **Gnomes in the remaining four houses:** 12 gnomes
- * **Number of remaining houses:** 4 houses
- * **Gnomes per house (if evenly distributed):** $12 \text{ gnomes} / 4 \text{ houses} = 3 \text{ gnomes/house}$

Therefore, the math is correct for both verification questions.

[Final Answer]:

The fifth house has 8 gnomes.

11. TREE OF THOUGHTS:

[The 3 Branches]:

Here are 3 completely different step-by-step ways to solve the problem:

Solution 1: Direct Calculation of Gnomes on the First Four Houses

Step 1: Calculate the total number of gnomes on the first four houses.

- * Each of the first four houses has 3 gnomes.
- * So, the total gnomes on these houses is $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$.

Step 2: Subtract the gnomes on the first four houses from the total number of gnomes.

- * The total number of gnomes on the street is 20.
- * The number of gnomes on the first four houses is 12.
- * Therefore, the number of gnomes on the fifth house is $20 \text{ total gnomes} - 12 \text{ gnomes} = 8 \text{ gnomes}$.

Answer: The fifth house has 8 gnomes.

Solution 2: Thinking About the "Missing" Gnomes

Step 1: Determine how many gnomes would be on the first four houses if they all had the same number as the fifth house (which we don't know yet).

- * This step is a bit of a conceptual setup. We know the total is 20 and there are 5 houses. If all houses had an equal number of gnomes, each would have $20 / 5 = 4$ gnomes. However, we are given specific information about the first four houses.

****Step 2: Focus on the gnomes that *are* accounted for on the first four houses.****

* We are told the first four houses *each* have 3 gnomes.

****Step 3: Calculate how many gnomes are *left over* after accounting for the gnomes on the first four houses.****

* The first four houses contribute 4 houses * 3 gnomes/house = 12 gnomes to the total.

* The total number of gnomes is 20.

* The remaining gnomes must be on the fifth house: 20 total gnomes - 12 gnomes = 8 gnomes.

****Answer:**** The fifth house has 8 gnomes.

Solution 3: Using an Equation

****Step 1: Define a variable for the unknown quantity.****

* Let 'x' represent the number of gnomes on the fifth house.

****Step 2: Set up an equation that represents the total number of gnomes.****

* The total number of gnomes is the sum of gnomes on each of the five houses.

* Gnomes on house 1 = 3

* Gnomes on house 2 = 3

* Gnomes on house 3 = 3

* Gnomes on house 4 = 3

* Gnomes on house 5 = x

* The total is 20.

* So, the equation is: $3 + 3 + 3 + 3 + x = 20$

****Step 3: Simplify and solve the equation for 'x'.****

* Combine the known numbers: $12 + x = 20$

* Subtract 12 from both sides of the equation to isolate 'x':

$$12 + x - 12 = 20 - 12$$

$$x = 8$$

****Answer:**** The fifth house has 8 gnomes.

[Evaluator Final Answer]:

8

--- QUESTION 2 ---

9. REFLEXION:

[Draft]: + Overtime pay

* ****Total earnings:**** $\$720.00 + \$270.00 = \$990.00$

****Therefore, Tina makes \$990.00.**...**

[Critique]:

Here's a critique of the draft answer:

****Overall Assessment:****

The math in the draft answer is ****correct****. The logic for calculating regular pay, overtime pay, and total earnings is sound and follows the problem's conditions.

****Specific Points and Potential Improvements:****

* ****Clarity of "per shift" vs. "per day":**** The problem states "per shift," but the answer consistently uses "per day." While in this specific scenario, working 10 hours **every day** for 5 days means each shift is also a day, it's worth noting this slight ambiguity. If Tina had multiple shifts in a day, the calculation would be different. However, given the phrasing, assuming one shift per day is reasonable.

* ****Redundancy in "Total regular pay for 5 days":**** The calculation is correct, but the phrasing could be slightly more direct. Instead of calculating daily regular pay and then multiplying by 5, you could directly calculate the total regular hours and then multiply by the regular rate.

* ****Redundancy in "Total overtime pay for 5 days":**** Similar to the regular pay, the calculation is correct, but the phrasing could be more direct.

****Revised Approach (for slightly more conciseness):****

Here's how to break down Tina's earnings:

****1. Calculate her regular pay:****

- * ****Regular hours per day:**** 8 hours
- * ****Regular pay per hour:**** \$18.00
- * ****Total regular hours for 5 days:**** 8 hours/day * 5 days = 40 hours
- * ****Total regular pay:**** 40 hours * \$18.00/hour = \$720.00

****2. Calculate her overtime pay:****

- * ****Overtime hours per day:**** 10 hours (total) - 8 hours (regular) = 2 hours
- * ****Overtime hourly rate:**** \$18.00 (regular wage) + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00
- * ****Total overtime hours for 5 days:**** 2 hours/day * 5 days = 10 hours
- * ****Total overtime pay:**** 10 hours * \$27.00/hour = \$270.00

****3. Calculate her total earnings:****

- * **Total earnings:** Total regular pay + Total overtime pay
- * **Total earnings:** \$720.00 + \$270.00 = \$990.00

Therefore, Tina makes \$990.00.

Conclusion:

The original draft answer is **mathematically correct**. The critique focuses on minor stylistic improvements for conciseness, but the core calculations and the final answer are accurate.

[Final Revision]:

Here's how to break down Tina's earnings:

1. Calculate her regular pay:

- * **Regular hours per day:** 8 hours
- * **Regular pay per hour:** \$18.00
- * **Total regular hours for 5 days:** 8 hours/day * 5 days = 40 hours
- * **Total regular pay:** 40 hours * \$18.00/hour = \$720.00

2. Calculate her overtime pay:

- * **Overtime hours per day:** 10 hours (total) - 8 hours (regular) = 2 hours
- * **Overtime hourly rate:** \$18.00 (regular wage) + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00
- * **Total overtime hours for 5 days:** 2 hours/day * 5 days = 10 hours
- * **Total overtime pay:** 10 hours * \$27.00/hour = \$270.00

3. Calculate her total earnings:

- * **Total earnings:** Total regular pay + Total overtime pay
- * **Total earnings:** \$720.00 + \$270.00 = \$990.00

Therefore, Tina makes \$990.00.

10. CHAIN-OF-VERIFICATION:

[Baseline Draft]: days

- * **Total earnings:** \$198.00/day * 5 days = \$990.00

Answer: Tina makes \$990.00 in 5 days....

[Verification Plan]:

Here are two verification questions to check if Tina's earnings calculation is correct:

1. ****If Tina worked exactly 8 hours each day for 5 days, what would her total earnings be?****
(This checks if the regular pay calculation is correct and if the overtime calculation is being added appropriately.)

2. ****If Tina worked 10 hours each day for 5 days, and her overtime rate was double her regular rate (\$36/hour), what would her total earnings be?**** (This checks if the overtime rate calculation is correct and if the total daily pay is being calculated and then multiplied by the number of days.)

[Verifications Executed]:

Let's break down Tina's earnings based on the information provided in the questions. We'll need to make some assumptions about her regular hourly rate since it's not explicitly stated in the questions themselves, but we can infer it from the overtime rate in question 2.

****Assumption:**** From question 2, we know her overtime rate is double her regular rate and is \$36/hour. Therefore, her ****regular hourly rate is \$18/hour**** ($\$36 / 2 = \18).

Now, let's answer the verification questions:

****1. If Tina worked exactly 8 hours each day for 5 days, what would her total earnings be?****

* ****Regular hours per day:**** 8 hours

* ****Number of days:**** 5 days

* ****Regular hourly rate:**** \$18/hour

* ****Total regular hours worked:**** 8 hours/day * 5 days = 40 hours

* ****Total earnings:**** 40 hours * \$18/hour = ****\$720****

****Verification Check:**** This calculation confirms that the regular pay is calculated correctly by multiplying the total regular hours by the regular hourly rate. Since no overtime is worked in this scenario, the overtime calculation is not a factor, and the total earnings are solely based on regular pay.

****2. If Tina worked 10 hours each day for 5 days, and her overtime rate was double her regular rate (\$36/hour), what would her total earnings be?****

* ****Hours worked per day:**** 10 hours

* ****Number of days:**** 5 days

* ****Regular hourly rate:**** \$18/hour (inferred from overtime rate)

* ****Overtime rate:**** \$36/hour

****First, let's calculate the daily earnings:****

* ****Regular hours per day:**** 8 hours (assuming a standard 8-hour workday for overtime calculation)

* ****Overtime hours per day:**** 10 hours - 8 hours = 2 hours

* ****Daily regular pay:**** 8 hours * \$18/hour = \$144

* ****Daily overtime pay:**** 2 hours * \$36/hour = \$72

* ****Total daily earnings:**** \$144 + \$72 = \$216

****Now, let's calculate the total earnings for 5 days:****

* ****Total earnings:**** \$216/day * 5 days = ****\$1080****

****Verification Check:**** This calculation confirms that:

- * The overtime hours are correctly identified (hours exceeding the regular workday).
- * The overtime rate is correctly applied to the overtime hours.
- * The total daily pay is calculated by summing the regular pay and overtime pay for that day.
- * The total earnings are then correctly calculated by multiplying the total daily pay by the

number of days worked.

[Final Answer]:

The original calculation for Tina's earnings is correct.

****Original Calculation Summary:****

* ****Regular pay per day:**** \$144.00

* ****Overtime pay per day:**** \$54.00

* ****Total pay per day:**** \$198.00

* ****Total earnings for 5 days:**** \$990.00

****Verification 1:**** If Tina worked exactly 8 hours each day for 5 days, her total earnings would be \$720. This aligns with the regular pay calculation.

****Verification 2:**** If Tina worked 10 hours each day for 5 days, with an overtime rate of \$36/hour (implying a regular rate of \$18/hour), her total earnings would be \$1080. This also aligns with the overtime calculation.

****Final Verified Answer:**** Tina makes \$990.00 in 5 days.

11. TREE OF THOUGHTS:

[The 3 Branches]:

Here are three completely different step-by-step ways to solve Tina's earnings problem:

Way 1: Calculate Regular and Overtime Hours Separately

This method breaks down the total hours into regular and overtime components and then calculates the pay for each.

****Step 1: Determine the number of regular hours per shift.****

Tina is eligible for overtime if she works **more than** 8 hours. This means the first 8 hours of each shift are regular hours.

****Step 2: Determine the number of overtime hours per shift.****

If Tina works 10 hours per shift, and the first 8 are regular, then:

Overtime hours per shift = Total hours per shift - Regular hours per shift

Overtime hours per shift = 10 hours - 8 hours = 2 hours

****Step 3: Calculate the regular pay per shift.****

Regular pay per shift = Regular hours per shift * Hourly wage

Regular pay per shift = 8 hours * \$18.00/hour = \$144.00

****Step 4: Calculate the overtime hourly wage.****

Overtime hourly wage = Hourly wage + (1/2 * Hourly wage)

Overtime hourly wage = \$18.00 + (0.5 * \$18.00)

Overtime hourly wage = \$18.00 + \$9.00 = \$27.00

****Step 5: Calculate the overtime pay per shift.****

Overtime pay per shift = Overtime hours per shift * Overtime hourly wage

Overtime pay per shift = 2 hours * \$27.00/hour = \$54.00

****Step 6: Calculate the total pay per shift.****

Total pay per shift = Regular pay per shift + Overtime pay per shift

Total pay per shift = \$144.00 + \$54.00 = \$198.00

****Step 7: Calculate the total earnings for the week.****

Total earnings = Total pay per shift * Number of days worked

Total earnings = \$198.00/day * 5 days = \$990.00

Way 2: Calculate Total Earnings by Combining Regular and Overtime Rates for Each Shift

This method calculates the total earnings for a single shift first, then multiplies by the number of days.

****Step 1: Determine the number of regular and overtime hours within one 10-hour shift.****

Regular hours = 8 hours

Overtime hours = 10 hours - 8 hours = 2 hours

****Step 2: Calculate the pay for the regular hours in one shift.****

Regular pay per shift = 8 hours * \$18.00/hour = \$144.00

****Step 3: Calculate the overtime rate.****

Overtime rate = \$18.00 * 1.5 = \$27.00/hour

****Step 4: Calculate the pay for the overtime hours in one shift.****

Overtime pay per shift = 2 hours * \$27.00/hour = \$54.00

****Step 5: Calculate the total earnings for one 10-hour shift.****

Total earnings per shift = Regular pay per shift + Overtime pay per shift

Total earnings per shift = \$144.00 + \$54.00 = \$198.00

****Step 6: Calculate the total earnings for 5 days.****

Total earnings = Total earnings per shift * Number of days worked

Total earnings = \$198.00/day * 5 days = \$990.00

Way 3: Calculate Total Hours and Apply Weighted Average

This method calculates the total hours worked and then uses a weighted average to account for the different pay rates.

****Step 1: Calculate the total hours worked over 5 days.****

Total hours worked = Hours per day * Number of days worked

Total hours worked = 10 hours/day * 5 days = 50 hours

****Step 2: Determine the proportion of regular and overtime hours worked.****

For every 10-hour shift, there are 8 regular hours and 2 overtime hours.

Proportion of regular hours per shift = 8 hours / 10 hours = 0.8

Proportion of overtime hours per shift = 2 hours / 10 hours = 0.2

****Step 3: Calculate the total number of regular hours worked over 5 days.****

Total regular hours = Total hours worked * Proportion of regular hours per shift

Total regular hours = 50 hours * 0.8 = 40 hours

****Step 4: Calculate the total number of overtime hours worked over 5 days.****

Total overtime hours = Total hours worked * Proportion of overtime hours per shift

Total overtime hours = 50 hours * 0.2 = 10 hours

****Step 5: Calculate the overtime hourly wage.****

Overtime hourly wage = \$18.00 + (\$18.00 / 2) = \$18.00 + \$9.00 = \$27.00

****Step 6: Calculate the total earnings from regular hours.****

Regular earnings = Total regular hours * Hourly wage

Regular earnings = 40 hours * \$18.00/hour = \$720.00

****Step 7: Calculate the total earnings from overtime hours.****

Overtime earnings = Total overtime hours * Overtime hourly wage

Overtime earnings = 10 hours * \$27.00/hour = \$270.00

****Step 8: Calculate the total earnings.****

Total earnings = Regular earnings + Overtime earnings

Total earnings = \$720.00 + \$270.00 = \$990.00

[Evaluator Final Answer]:

\$990.00

--- QUESTION 3 ---

9. REFLEXION:

[Draft]: nce = 840 pieces - 825 pieces = 15 pieces

****Answer:**** Anna gets 15 more pieces of candy than Billy....

[Critique]:

Here's a critique of the provided answer:

****Critique:****

The math in the draft answer is ****correct****. All calculations are performed accurately, and the steps logically lead to the correct conclusion.

****No errors were found.****

****Explanation of why the math is correct:****

* ****Anna's total candy:**** 14 pieces/house * 60 houses = 840 pieces. This multiplication is correct.

* ****Billy's total candy:**** 11 pieces/house * 75 houses = 825 pieces. This multiplication is correct.

* ****Difference:**** 840 pieces - 825 pieces = 15 pieces. This subtraction is correct.

The answer clearly breaks down the problem into manageable steps and presents the final answer in a straightforward manner.

[Final Revision]:

Here's how to solve the problem step-by-step:

****1. Calculate Anna's total candy:****

* Anna gets 14 pieces of candy per house.

* There are 60 houses in her subdivision.

* Total candy for Anna = 14 pieces/house * 60 houses = 840 pieces

****2. Calculate Billy's total candy:****

* Billy gets 11 pieces of candy per house.

* There are 75 houses in his subdivision.

* Total candy for Billy = 11 pieces/house * 75 houses = 825 pieces

****3. Find the difference:****

* To find out how many more pieces Anna gets, subtract Billy's total from Anna's total.

* Difference = Anna's candy - Billy's candy

* Difference = 840 pieces - 825 pieces = 15 pieces

****Answer:**** Anna gets 15 more pieces of candy than Billy.

10. CHAIN-OF-VERIFICATION:

[Baseline Draft]: $840 \text{ pieces} - 825 \text{ pieces} = 15 \text{ pieces}$

****Answer:**** Anna gets 15 more pieces of candy than Billy....

[Verification Plan]:

Here are two verification questions to check if the math is correct:

1. ****If Anna had collected 10 pieces of candy per house for 60 houses, and Billy collected 10 pieces of candy per house for 75 houses, would Anna still have more candy?**** (This checks the understanding of how the number of houses impacts the total and if the initial calculation of Anna having more is robust.)

2. ****If Anna collected 14 pieces of candy from 75 houses and Billy collected 11 pieces of candy from 60 houses, who would have more candy and by how much?**** (This swaps the number of houses to see if the calculation logic holds when the roles are reversed and the numbers are different.)

[Verifications Executed]:

Let's break down these verification questions step-by-step:

****Question 1: If Anna had collected 10 pieces of candy per house for 60 houses, and Billy collected 10 pieces of candy per house for 75 houses, would Anna still have more candy?****

* ****Anna's candy:**** 10 pieces/house * 60 houses = 600 pieces of candy

* ****Billy's candy:**** 10 pieces/house * 75 houses = 750 pieces of candy

****Answer to Question 1:**** No, Anna would ****not**** still have more candy. In this scenario, Billy would have more candy.

****Question 2: If Anna collected 14 pieces of candy from 75 houses and Billy collected 11 pieces of candy from 60 houses, who would have more candy and by how much?****

- * ****Anna's candy:**** 14 pieces/house * 75 houses = 1050 pieces of candy
- * ****Billy's candy:**** 11 pieces/house * 60 houses = 660 pieces of candy

To find out who has more and by how much:

- * Anna has 1050 pieces.
- * Billy has 660 pieces.
- * Difference: $1050 - 660 = 390$ pieces

****Answer to Question 2:**** Anna would have more candy, and she would have ****390**** more pieces than Billy.

[Final Answer]:

The original problem states:

- * Anna gets 14 pieces of candy per house and visits 60 houses.
- * Billy gets 11 pieces of candy per house and visits 75 houses.

The original solution correctly calculated:

- * Anna's total candy: $14 * 60 = 840$ pieces
- * Billy's total candy: $11 * 75 = 825$ pieces
- * The difference: $840 - 825 = 15$ pieces

The original answer states: "Anna gets 15 more pieces of candy than Billy."

The verifications provided are:

****Verification 1:****

- * Anna: 10 pieces/house * 60 houses = 600 pieces
- * Billy: 10 pieces/house * 75 houses = 750 pieces
- * Result: Billy has more candy. This verification is correct.

****Verification 2:****

- * Anna: 14 pieces/house * 75 houses = 1050 pieces
- * Billy: 11 pieces/house * 60 houses = 660 pieces
- * Difference: $1050 - 660 = 390$ pieces
- * Result: Anna has 390 more pieces. This verification is correct.

All calculations and conclusions in the original problem and its verifications are accurate.

****Final Verified Answer:**** Anna gets 15 more pieces of candy than Billy.

11. TREE OF THOUGHTS:

[The 3 Branches]:

Here are 3 completely different step-by-step ways to solve the problem:

Method 1: Calculate Total Candy for Each Person First

This method focuses on finding the total candy each person collects before comparing them.

****Step 1: Calculate Anna's total candy.****

* Anna gets 14 pieces of candy per house and visits 60 houses.

* Total candy for Anna = (pieces per house) $\times$ (number of houses)

* Total candy for Anna = 14×60

****Step 2: Perform the multiplication for Anna's candy.****

* $14 \times 60 = 840$ pieces of candy.

****Step 3: Calculate Billy's total candy.****

* Billy gets 11 pieces of candy per house and visits 75 houses.

* Total candy for Billy = (pieces per house) $\times$ (number of houses)

* Total candy for Billy = 11×75

****Step 4: Perform the multiplication for Billy's candy.****

* $11 \times 75 = 825$ pieces of candy.

****Step 5: Find the difference in candy collected.****

* To find out how many more pieces of candy Anna gets, subtract Billy's total from Anna's total.

* Difference = Anna's total candy - Billy's total candy

* Difference = $840 - 825$

****Step 6: Perform the subtraction to find the final answer.****

* $840 - 825 = 15$ pieces of candy.

****Answer:**** Anna gets 15 more pieces of candy.

Method 2: Calculate the Difference in Candy Per House and Then Total the Difference

This method focuses on the difference in candy received per house and then scales that difference up based on the number of houses.

****Step 1: Find the difference in candy per house.****

* Anna gets 14 pieces per house, and Billy gets 11 pieces per house.

* Difference per house = Anna's pieces per house - Billy's pieces per house

* Difference per house = $14 - 11$

****Step 2: Perform the subtraction for the difference per house.****

* $14 - 11 = 3$ pieces of candy more per house for Anna.

****Step 3: Calculate the *additional* candy Anna gets from her houses compared to Billy if he also visited her subdivision.****

* Anna visits 60 houses, and gets 3 more pieces per house than Billy.

* Additional candy from Anna's houses = (difference per house) $\times$ (Anna's number of houses)

* Additional candy from Anna's houses = 3×60

****Step 4: Perform the multiplication for this additional candy.****

* $3 \times 60 = 180$ pieces of candy.

****Step 5: Calculate the candy Billy gets from his houses compared to Anna if she also visited his subdivision.****

* Billy visits 75 houses, and gets 3 fewer pieces per house than Anna.

* Fewer pieces from Billy's houses = (difference per house) $\times$ (Billy's number of houses)

* Fewer pieces from Billy's houses = 3×75

****Step 6: Perform the multiplication for these fewer pieces.****

* $3 \times 75 = 225$ pieces of candy.

****Step 7: Calculate the net difference in candy.****

* Anna got 180 more pieces from her houses.

* Billy got 225 fewer pieces from his houses than Anna would have if she went there.

* Net difference = Additional candy from Anna's houses - Fewer pieces from Billy's houses

* Net difference = $180 - 225$

****Step 8: Perform the subtraction to find the final answer.****

* $180 - 225 = -45$. This means Billy got 45 more pieces of candy.

* ***Correction:** The question asks how many ***more*** pieces Anna gets. Since the calculation resulted in a negative number for Anna, it means Billy got more. We need to find the absolute difference and state who got more.

* The difference in total candy is the absolute value of -45 , which is 45.

* Since the calculation showed Billy got more, Anna got 45 fewer pieces.

* Therefore, Anna gets ****15**** more pieces of candy. (Wait, this is incorrect. Let's rethink Step 7 and 8 for clarity).

****Revised Step 7: Calculate the net difference in candy by comparing the *total difference* in candy from each subdivision.****

* Anna gets 3 more pieces per house for 60 houses. This is a gain of $3 \times 60 = 180$ pieces for Anna.

* Billy gets 3 fewer pieces per house for 75 houses. This is a loss of $3 \times 75 = 225$ pieces for Anna (or a gain of 225 for Billy).

* To find how many more Anna got, we can think of it as her gains minus his gains (or her losses).

* Anna's net gain = (Anna's gain per house * Anna's houses) - (Billy's gain per house * Billy's houses)

* Anna's net gain = $(3 * 60) - (3 * 75)$

* Anna's net gain = $180 - 225$

Revised Step 8: Perform the subtraction to find the final answer.

* $180 - 225 = -45$.

* A negative result means Anna got *fewer* pieces of candy than Billy.

* The absolute difference is $|-45| = 45$.

* This means Billy got 45 more pieces than Anna.

* The question asks how many *more* pieces Anna gets. Since the result is negative, Anna gets **-45** more pieces, which is the same as getting 45 fewer pieces.

* **Let's re-evaluate the question and the core difference.** The core difference is *how many more pieces Anna gets*.

**Let's simplify Method 2's approach to focus on the *total difference*.

Revised Method 2 (Simplified):

Step 1: Calculate Anna's total candy.

* $14 \text{ pieces/house} * 60 \text{ houses} = 840 \text{ pieces}$

Step 2: Calculate Billy's total candy.

* $11 \text{ pieces/house} * 75 \text{ houses} = 825 \text{ pieces}$

Step 3: Find the difference directly.

* Anna's total - Billy's total = $840 - 825 = 15 \text{ pieces}$.

Answer: Anna gets 15 more pieces of candy.

(Note: The initial attempt at Method 2 was overly complicated and led to confusion. The simplified approach is essentially Method 1 but framed slightly differently in its thought process.)

Method 3: Focus on the "Extra" Candy for Anna and the "Shortfall" for Billy

This method breaks down the candy collection into components that directly address the "how many more" question.

Step 1: Determine how many more pieces Anna gets per house.

* Anna gets 14 pieces per house.

* Billy gets 11 pieces per house.

* The difference is $14 - 11 = 3$ more pieces for Anna per house.

****Step 2: Calculate the total "extra" candy Anna gets from her houses.****

* Anna visits 60 houses.

* Total extra candy for Anna = (extra pieces per house) × (Anna's number of houses)

* Total extra candy for Anna = $3 \times 60 = 180$ pieces.

****Step 3: Determine how many fewer pieces Billy gets per house compared to Anna.****

* This is the same difference as in Step 1: 3 fewer pieces for Billy per house.

****Step 4: Calculate the "shortfall" in candy for Billy across his houses.****

* Billy visits 75 houses.

* Total shortfall for Billy = (fewer pieces per house) × (Billy's number of houses)

* Total shortfall for Billy = $3 \times 75 = 225$ pieces.

****Step 5: Compare Anna's "extra" to Billy's "shortfall" to find the net difference.****

* Anna has an excess of 180 pieces from her subdivision.

* Billy has a deficit of 225 pieces compared to Anna's rate in his subdivision.

* To find how many *more* Anna gets, we can think of it as: Anna's gains - Billy's gains (from Anna's perspective).

* Anna's net gain = (Anna's extra from her houses) - (Billy's shortfall from his houses)

* Anna's net gain = $180 - 225$

****Step 6: Calculate the final difference.****

* $180 - 225 = -45$.

* This means Anna has 45 *fewer* pieces than Billy.

* Therefore, the question "how many more pieces of candy does Anna get?" means we need to find the difference.

* If Anna gets -45 more, it means she gets 45 fewer.

* ****Let's re-frame the comparison to be more direct.****

****Revised Method 3 (Focus on Direct Comparison):****

****Step 1: Calculate the total candy Anna collects.****

* Anna's total = 14 pieces/house * 60 houses = 840 pieces

****Step 2: Calculate the total candy Billy collects.****

* Billy's total = 11 pieces/house * 75 houses = 825 pieces

****Step 3: Find the difference between Anna's total and Billy's total.****

* Difference = Anna's total - Billy's total

* Difference = $840 - 825$

****Step 4: Perform the subtraction.****

* $840 - 825 = 15$ pieces.

****Answer:**** Anna gets 15 more pieces of candy.

(Similar to Method 2, the initial thought process for Method 3 could be complex. The most straightforward way to answer "how many more" is to calculate each total and then subtract.)

****Conclusion on the methods:**** While the prompt asks for 3 *completely different* step-by-step ways, in practice, for this specific problem, the most fundamental and clear approaches will revolve around calculating individual totals first. The variations come in how you break down the calculations or the intermediate steps. The core mathematical operations remain multiplication and subtraction. The key is to ensure each method arrives at the correct answer through distinct logical pathways.

[Evaluator Final Answer]:

15

--- QUESTION 1 ---

12. MULTI-AGENT DEBATE:

[Agent 1]:

Here's how to solve the problem:

1. ****Gnomes in the first four houses:**** $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$
2. ****Gnomes in the fifth house:**** $20 \text{ total gnomes} - 12 \text{ gnomes} = 8 \text{ gnomes}$

The fifth house has ****8**** gnomes.

[Agent 2]:

Here's how to solve this problem differently, focusing on the total and then finding the remainder:

****The "Subtraction from the Total" Method****

1. ****Identify the total number of gnomes:**** The problem states there are a total of 20 gnomes on the street.
2. ****Calculate gnomes in the first four houses:****
 - * Each of the first four houses has 3 gnomes.
 - * So, the first four houses have $4 \text{ houses} * 3 \text{ gnomes/house} = 12 \text{ gnomes}$.
3. ****Find the gnomes in the fifth house by subtracting:****
 - * Subtract the gnomes in the first four houses from the total number of gnomes.
 - * $20 \text{ total gnomes} - 12 \text{ gnomes in the first four houses} = 8 \text{ gnomes}$.

****Therefore, the fifth house has 8 gnomes.****

[Judge Final Verdict]:

Both Agent 1 and Agent 2 are correct. They both arrive at the same answer using the same logical steps.

The final answer is $\boxed{8}$.

14. REACT Final Answer:

The problem states that there are 5 houses on a street.

The first four houses each have 3 gnomes.

The total number of gnomes on the street is 20.

We need to find the number of gnomes in the fifth house.

First, calculate the total number of gnomes in the first four houses.

Number of gnomes per house = 3

Number of houses = 4

Total gnomes in the first four houses = Number of gnomes per house $\times$ Number of houses

Total gnomes in the first four houses = $3 \times 4 = 12$

The observation provided in the history states: "There are 15 gnomes in the first four houses."

Let's re-evaluate the calculation based on the problem statement.

Number of gnomes in the first house = 3

Number of gnomes in the second house = 3

Number of gnomes in the third house = 3

Number of gnomes in the fourth house = 3

Total gnomes in the first four houses = $3 + 3 + 3 + 3 = 12$.

There seems to be a discrepancy between the calculation based on the problem statement (12 gnomes) and the observation provided in the history (15 gnomes). We will proceed with the calculation based on the problem statement as it is the primary source of information.

Total number of gnomes on the street = 20.

Total number of gnomes in the first four houses = 12.

To find the number of gnomes in the fifth house, subtract the total number of gnomes in the first four houses from the total number of gnomes on the street.

Number of gnomes in the fifth house = Total gnomes on the street - Total gnomes in the first four houses

Number of gnomes in the fifth house = $20 - 12 = 8$.

If we were to use the observation from the history:

Total number of gnomes on the street = 20.

Total number of gnomes in the first four houses (from observation) = 15.

Number of gnomes in the fifth house = $20 - 15 = 5$.

However, the problem statement clearly says "each of the first four houses has 3 gnomes". This leads to 12 gnomes. The observation in the history seems to be incorrect based on the problem statement. We will follow the problem statement.

Number of gnomes in the first four houses = 4 houses * 3 gnomes/house = 12 gnomes.

Total gnomes on the street = 20 gnomes.

Number of gnomes in the fifth house = Total gnomes - Gnomes in the first four houses

Number of gnomes in the fifth house = $20 - 12 = 8$ gnomes.

The final answer is $\boxed{8}$.

--- QUESTION 2 ---

12. MULTI-AGENT DEBATE:

[Agent 1]:

Here's how to solve this problem step-by-step:

****1. Calculate the overtime rate:****

* Tina's regular hourly wage: \$18.00

* Overtime rate is regular wage + 1/2 regular wage: $\$18.00 + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00$ per hour

****2. Determine regular hours and overtime hours per day:****

* Tina works 10 hours per day.

* Regular hours per day: 8 hours

* Overtime hours per day: 10 hours - 8 hours = 2 hours

****3. Calculate earnings for regular hours per day:****

* Regular earnings per day: 8 hours * \$18.00/hour = \$144.00

****4. Calculate earnings for overtime hours per day:****

* Overtime earnings per day: 2 hours * \$27.00/hour = \$54.00

****5. Calculate total earnings per day:****

* Total earnings per day: \$144.00 (regular) + \$54.00 (overtime) = \$198.00

****6. Calculate total earnings for 5 days:****

* Total earnings for 5 days: \$198.00/day * 5 days = \$990.00

****Therefore, Tina makes \$990.00 if she works 10 hours every day for 5 days.****

[Agent 2]:

Here's a different way to solve Tina's earnings problem, focusing on breaking down the overtime calculation:

****1. Calculate the Overtime Rate:****

* ****Regular hourly wage:**** \$18.00

* ****Overtime premium:**** $1/2$ of her hourly wage = $\$18.00 / 2 = \9.00

* ****Total overtime hourly rate:**** Regular wage + Overtime premium = $\$18.00 + \$9.00 = \$27.00$

****2. Determine Overtime Hours per Day:****

* ****Total hours worked per day:**** 10 hours

* ****Regular hours per shift:**** 8 hours

* ****Overtime hours per day:**** Total hours - Regular hours = 10 hours - 8 hours = 2 hours

****3. Calculate Daily Earnings:****

* ****Earnings from regular hours:**** 8 hours * \$18.00/hour = \$144.00

* ****Earnings from overtime hours:**** 2 hours * \$27.00/hour = \$54.00

* ****Total earnings per day:**** \$144.00 + \$54.00 = \$198.00

****4. Calculate Total Earnings for the Week:****

* ****Number of days worked:**** 5 days

* ****Total earnings:**** \$198.00/day * 5 days = \$990.00

****Therefore, Tina makes \$990.00.****

****Key differences in this approach:****

* ****Explicitly calculating the overtime rate first:**** This clearly separates the calculation of the higher pay rate for overtime hours.

* ****Breaking down daily earnings:**** Instead of directly calculating total regular and overtime hours for the week, this method calculates the earnings for each day first, then multiplies by the number of days. This can make the logic more digestible for some.

* ****Clearer labeling of each step:**** Using descriptive headings for each calculation makes the process easier to follow.

[Judge Final Verdict]:

Both agents arrive at the correct answer, \$990.00. Their methods are sound and lead to the same result.

The final answer is $\boxed{\$990.00}$.

14. REACT Final Answer:

Tina's regular hourly wage is \$18.00.

Overtime is paid at her hourly wage + 1/2 her hourly wage.

Overtime rate = $\$18.00 + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00$ per hour.

Tina works 10 hours per day for 5 days.

For each day, the first 8 hours are regular hours, and the remaining hours are overtime hours.

Regular hours per day = 8 hours.

Overtime hours per day = 10 hours - 8 hours = 2 hours.

Total regular hours for 5 days = 8 hours/day * 5 days = 40 hours.

Total overtime hours for 5 days = 2 hours/day * 5 days = 10 hours.

Earnings from regular hours = Total regular hours * Regular hourly wage

Earnings from regular hours = 40 hours * \$18.00/hour = \$720.00.

Earnings from overtime hours = Total overtime hours * Overtime hourly wage

Earnings from overtime hours = 10 hours * \$27.00/hour = \$270.00.

Total money Tina makes = Earnings from regular hours + Earnings from overtime hours

Total money Tina makes = \$720.00 + \$270.00 = \$990.00.

The final answer is $\boxed{\$990.00}$.

--- QUESTION 3 ---

12. MULTI-AGENT DEBATE:

[Agent 1]:

Here's how to solve the problem step-by-step:

****1. Calculate Anna's total candy:****

* Anna gets 14 pieces of candy per house.

* There are 60 houses in her subdivision.

* Anna's total candy = 14 pieces/house * 60 houses = 840 pieces of candy

****2. Calculate Billy's total candy:****

* Billy gets 11 pieces of candy per house.

* There are 75 houses in his subdivision.

* Billy's total candy = 11 pieces/house * 75 houses = 825 pieces of candy

****3. Find the difference:****

- * To find out how many more pieces Anna gets, subtract Billy's candy from Anna's candy.
- * $\text{Difference} = \text{Anna's candy} - \text{Billy's candy} = 840 - 825 = 15$ pieces of candy

****Answer:**** Anna gets 15 more pieces of candy than Billy.

[Agent 2]:

Here's a different way to solve the problem, focusing on the *difference in candy per house* and then applying that to the *number of houses*.

****1. Find the difference in candy per house:****

- * Anna gets 14 pieces per house.
- * Billy gets 11 pieces per house.
- * The difference is $14 - 11 = 3$ more pieces per house for Anna.

****2. Determine how many houses contribute to this difference:****

- * Anna's subdivision has 60 houses.
- * Billy's subdivision has 75 houses.

This is where it gets a little tricky to apply the "difference per house" directly because the number of houses is different. We need to consider the scenarios separately for a moment to ensure we're comparing apples to apples.

****Alternative Approach - Focusing on the Total Candy and then Subtracting:****

This is the most straightforward "different" approach that still accurately solves the problem.

****1. Calculate Anna's total candy:****

- * Anna gets 14 pieces per house.
- * There are 60 houses in her subdivision.
- * Anna's total candy = $14 \text{ pieces/house} * 60 \text{ houses} = 840$ pieces.

****2. Calculate Billy's total candy:****

- * Billy gets 11 pieces per house.
- * There are 75 houses in his subdivision.
- * Billy's total candy = $11 \text{ pieces/house} * 75 \text{ houses} = 825$ pieces.

****3. Find the difference:****

- * To find how many more pieces Anna gets, subtract Billy's total from Anna's total.
- * $\text{Difference} = \text{Anna's total candy} - \text{Billy's total candy}$
- * $\text{Difference} = 840 \text{ pieces} - 825 \text{ pieces} = 15$ pieces.

****Answer:**** Anna gets 15 more pieces of candy.

[Judge Final Verdict]:

Both agents arrive at the correct answer, but Agent 1's explanation is more direct and less confusing. Agent 2 attempts a different approach by looking at the difference in candy per house, but then correctly identifies that this approach becomes complicated due to the different number of houses and reverts to the standard method.

The core of the problem requires calculating the total candy for each person and then finding the difference.

****Agent 1's method is the most efficient and clear way to solve this problem.****

The final answer is $\boxed{15}$.

14. REACT Final Answer:

Anna's subdivision has 60 houses, and she gets 14 pieces of candy per house.

Total candy for Anna = Number of houses in Anna's subdivision * Candy per house

Total candy for Anna = 60 houses * 14 pieces/house = 840 pieces of candy.

Billy's subdivision has 75 houses, and he gets 11 pieces of candy per house.

Total candy for Billy = Number of houses in Billy's subdivision * Candy per house

Total candy for Billy = 75 houses * 11 pieces/house = 825 pieces of candy.

To find out how many more pieces of candy Anna gets, subtract Billy's total candy from Anna's total candy.

Difference = Total candy for Anna - Total candy for Billy

Difference = 840 pieces - 825 pieces = 15 pieces.

The final answer is $\boxed{15}$.

--- QUESTION 1 ---

15. DSPy:

[Reasoning Trace]:

First, calculate the total number of gnomes in the first four houses: 4 houses * 3 gnomes/house = 12 gnomes. Then, subtract this amount from the total number of gnomes on the street to find the number of gnomes in the fifth house: 20 total gnomes - 12 gnomes = 8 gnomes.

[Final Answer]:

8

--- QUESTION 2 ---

15. DSPy:

[Reasoning Trace]:

The problem asks to calculate Tina's total earnings for a week, considering her regular hourly wage and overtime pay.

First, identify Tina's regular hourly wage: \$18.00.

Next, determine the threshold for overtime: more than 8 hours per shift.

Calculate the overtime pay rate: regular hourly wage + 1/2 of the regular hourly wage.

Overtime rate = $\$18.00 + (1/2 * \$18.00) = \$18.00 + \$9.00 = \$27.00$.

Tina works 10 hours per day for 5 days.

For each day, Tina works 10 hours. The first 8 hours are regular hours, and the remaining 2 hours ($10 - 8 = 2$) are overtime hours.

Calculate the daily earnings:

Earnings from regular hours = 8 hours * \$18.00/hour = \$144.00.

Earnings from overtime hours = 2 hours * \$27.00/hour = \$54.00.

Total daily earnings = $\$144.00 + \$54.00 = \$198.00$.

Tina works for 5 days.

Total weekly earnings = Total daily earnings * 5 days = $\$198.00/\text{day} * 5 \text{ days}$.

Total weekly earnings = \$990.00.

Let's re-verify the steps.

Hourly wage = \$18.00

Overtime threshold = 8 hours

Overtime rate = $\$18.00 + 0.5 * \$18.00 = \$18.00 + \$9.00 = \$27.00$

Hours worked per day = 10 hours

Number of days worked = 5 days

For each day:

Regular hours = 8 hours

Overtime hours = $10 - 8 = 2$ hours

Daily earnings = (Regular hours * Hourly wage) + (Overtime hours * Overtime rate)

Daily earnings = $(8 * \$18.00) + (2 * \$27.00)$

Daily earnings = $\$144.00 + \54.00

Daily earnings = \$198.00

Total earnings for 5 days = Daily earnings * 5

Total earnings = $\$198.00 * 5$

Total earnings = \$990.00.

The calculations seem correct.

Alternative approach:

Total hours worked in 5 days = 10 hours/day * 5 days = 50 hours.

Total regular hours in 5 days = 8 hours/day * 5 days = 40 hours.

Total overtime hours in 5 days = $(10 - 8) \text{ hours/day} * 5 \text{ days} = 2 \text{ hours/day} * 5 \text{ days} = 10 \text{ hours}$.

Total earnings = (Total regular hours * Regular hourly wage) + (Total overtime hours * Overtime rate)

Total earnings = (40 hours * \$18.00/hour) + (10 hours * \$27.00/hour)

Total earnings = \$720.00 + \$270.00

Total earnings = \$990.00.

Both approaches yield the same result.

Final answer is \$990.00.

]]

[Final Answer]:

\$990.00

--- QUESTION 3 ---

15. DSPy:

[Reasoning Trace]:

The problem asks us to find the difference in the total number of candies collected by Anna and Billy.

First, we need to calculate the total number of candies Anna collected. Anna gets 14 pieces of candy per house and there are 60 houses in her subdivision. So, Anna's total candy = $14 * 60$.

Next, we need to calculate the total number of candies Billy collected. Billy gets 11 pieces of candy per house and there are 75 houses in his subdivision. So, Billy's total candy = $11 * 75$.

Finally, to find how many more pieces of candy Anna gets, we subtract Billy's total candy from Anna's total candy.

Anna's total candy = $14 * 60 = 840$.

Billy's total candy = $11 * 75 = 825$.

Difference = Anna's total candy - Billy's total candy = $840 - 825$.

Anna's total candy = 14 pieces/house * 60 houses = 840 pieces.

Billy's total candy = 11 pieces/house * 75 houses = 825 pieces.

The difference in the number of candies is Anna's candy - Billy's candy.

Difference = $840 - 825 = 15$.

Therefore, Anna gets 15 more pieces of candy than Billy.

[Final Answer]:

15

UPDATED PAL:

--- QUESTION 1 ---

13. PAL:

[Generated Python Code]:

total_gnomes = 20

houses_with_3_gnomes = 4

```
gnomes_per_house = 3
```

```
gnomes_in_first_four_houses = houses_with_3_gnomes * gnomes_per_house
```

```
gnomes_in_fifth_house = total_gnomes - gnomes_in_first_four_houses
```

```
print(gnomes_in_fifth_house)
```

[Executed Final Answer]:

8

--- QUESTION 2 ---

13. PAL:

[Generated Python Code]:

```
hourly_wage = 18.00
```

```
regular_hours_per_day = 8
```

```
overtime_rate_multiplier = 1.5
```

```
total_hours_per_day = 10
```

```
number_of_days = 5
```

```
regular_pay = hourly_wage * regular_hours_per_day
```

```
overtime_hours_per_day = total_hours_per_day - regular_hours_per_day
```

```
overtime_pay_rate = hourly_wage * overtime_rate_multiplier
```

```
overtime_pay_per_day = overtime_pay_rate * overtime_hours_per_day
```

```
daily_pay = regular_pay + overtime_pay_per_day
```

```
total_earnings = daily_pay * number_of_days
```

```
print(total_earnings)
```

[Executed Final Answer]:

990.0

--- QUESTION 3 ---

13. PAL:

[Generated Python Code]:

```
anna_candy_per_house = 14
```

```
anna_houses = 60
```

```
anna_total_candy = anna_candy_per_house * anna_houses
```

```
billy_candy_per_house = 11
```

```
billy_houses = 75
```

```
billy_total_candy = billy_candy_per_house * billy_houses
```

```
difference = anna_total_candy - billy_total_candy
```

```
print(difference)
```

[Executed Final Answer]:

15